Enterprise High-Level Architecture: C4 Context, Containers, Event Streaming & National DPI Integration

Governing Authority: Ministry of Cooperation, Government of India • *Theme: Sahakar Se Samriddhi*
Architectural Methodology: C4 Model • Asynchronous Event-Driven Microservices • **Version:** 1.0.0 (SIH 2026)

High-Level Enterprise Architecture: Seva Setu is designed as a sovereign, multi-tenant **Digital Public Infrastructure (DPI)**. It decouples consumer demand, cooperative labor supply, and institutional governance into autonomous, horizontally scalable layers. The architecture integrates C4 Level 1 (Enterprise Context), Level 2 (Container Topologies), asynchronous Redis/Kafka event streaming, Zero-Trust identity verification via Digilocker/UIDAI, PostGIS spatial clusters, and automated failover fallback state machines.

1. System Context Architecture (C4 Level 1: Enterprise Boundary)

Seva Setu establishes a trusted tripartite interface between the citizen, the decentralized cooperative labor force, and statutory public infrastructure. The platform coordinates transactions without extracting monopolistic commission margins.

External Entity / Actor	Interface Protocol	Information Flow & Strategic Integration Role
Citizens & NGOs	HTTPS / WSS / PWA	Submits service requests, selects triage urgency, receives unbundled itemized invoices, authorizes dual OTPs.
Cooperative Workers	WSS / Mobile REST API	Receives proximity broadcasts, visualizes guaranteed net pay, reports en-route hurdles, submits proof-of-work.
Ministry of Cooperation	HTTPS / Next.js Admin	Accesses national revenue audits, district-level workforce deficit heatmaps, and approves federation training grants.
Primary Societies (PACS)	Cooperative ERP Bridge	Performs in-person physical vetting of tools, handles dispute hearings, receives 1.5% statutory welfare cess.
UIDAI / Digilocker DPI	OAuth 2.0 / REST API	Performs zero-knowledge Aadhaar tokenization and fetches verifiable ITI / NSDC skill certifications.
Emergency ERSS 112	Encrypted Webhook API	Receives instant emergency SOS telemetry packets (live GPS, technician ID, vehicle plate) for police intervention.

2. Container Architecture (C4 Level 2: Subsystem Topology)

The container ecosystem is decoupled into edge routing, asynchronous micro-services, and dedicated persistence clusters:

- **Ingress & API Gateway:** An asynchronous FastAPI core fronted by an Nginx / Traefik load balancer. Handles rate limiting, JWT token validation, and request routing with non-blocking ASGI event loops.
- **Geospatial Dispatch Engine:** PostGIS-driven cluster coupled with an in-memory Redis 7 Geospatial cache (GEOSEARCH). Computes candidate proximity within 5km–15km radiuses in sub-18ms latency.
- **Decoupled 4-Layer Pricing Service:** Calculates unbundled transparent invoices, separating Worker Labor (100%), Transit Tariffs, Toolkit Allowances, 1.5% Cooperative Cess, and 3.5% Platform Tech Fees.
- **Zero-Downtime Fallback Coordinator:** Finite state machine engine that detects worker rejections or en-route vehicle hurdles, automatically cascading requests to the next nearest guild technician.
- **Material Procurement B2B Connector:** Mediates on-site spare part approvals between citizens, technicians, and local cooperative hardware stores, capturing a 3%–5% wholesale referral margin.
- **Persistence Clusters:** PostgreSQL 16 (Relational & Spatial), Redis 7 (Distributed Redlock & Caching), and S3/MinIO (Encrypted storage for proof-of-work photographs and trade credentials).

3. Core Subsystems & Operational Mechanisms

A. The Geospatial Triage Dispatch Subsystem

When an order is created, the spatial engine executes a two-stage filter: (1) *Spatial Radius Filter* using PostGIS ST_DWithin to find active certified technicians; (2) *Multi-Objective Scoring Function* sorting workers by distance, 14-day rolling reliability, customer ratings, and emergency certification bonuses. Critical P1 emergency requests bypass standard queues and route directly to technicians with 90%+ acceptance track records.

B. The Zero-Downtime Fallback State Machine

To ensure public service continuity, the fallback engine enforces two automated recovery protocols:
1. **Initial Broadcast Timeout:** Workers have 30 seconds to accept. If declined or timed out, the request escalates to the next nearest ranked technician within 15 seconds.
2. **En-Route Hurdle Protection:** If a traveling technician encounters an accident, flat tyre, or waterlogging, they tap *'Report Hurdle & Cancel'*. The system isolates the worker, logs the reason, and instantly re-dispatches the booking to the closest peer guild member. The citizen's service is never abandoned.

C. On-Site Dynamic Material Procurement Workflow

When on-site repairs require spare parts (e.g., replacement MCB switches, copper tubing):
• Technician uploads a live photograph of the damaged part with an itemized estimate.
• Citizen reviews and pre-authorizes the exact retail price in-app via digital escrow.
• The app generates a **Digital Procurement QR Pass** redeemable at verified cooperative hardware merchants.
• Merchant POS confirms part collection; citizen pays zero retail markup; platform retains a 4% B2B wholesale trade discount.

D. Emergency SOS Command & Telemetry Subsystem

Activated via a one-touch SOS button during domestic visits or driving gigs: Instantly packages an encrypted telemetry payload (live GPS coordinates, technician Aadhaar hash, vehicle registration, and active audio buffer) and triggers dual escalation to the 24/7 Command Center and local Police Control Room (ERSS 112 API).

E. Apex Ministry Cockpit & Deficit Heatmap Subsystem

Aggregates real-time municipal demand density against active technician supply across 15+ trade categories. Identifies localized workforce deficits (e.g., East Delhi electrical deficit >40%) and enables Ministry officials to deploy targeted vocational training cohorts and allocate PACS labor expansion grants with surgical precision.

4. Asynchronous Event-Driven Messaging Architecture

High-throughput events (live GPS tracking, radar HUD broadcasts, and fallback triggers) flow through an asynchronous event bus, preventing database locks during peak booking surges.

Event Channel / Topic	Publisher	Consumers / Subscribers	Action Triggered
booking.requested	Citizen PWA / Gateway	Spatial Dispatch Service	Executes PostGIS query and broadcasts job to candidate worker.
worker.declined	Worker Mobile App	Fallback Coordinator	Appends worker to rejection list; triggers auto-reassignment.
worker.hurdle_reported	Worker Mobile App	Fallback Engine & Admin Desk	Isolates failed technician; re-dispatches booking to guild peer.
booking.otp_verified	API Gateway	Ledger & Settlement Service	Starts service timer (START_OTP) or releases instant payout (END_OTP).
safety.sos_triggered	Citizen / Worker App	ERSS 112 API, Command Center	Dispatches encrypted telemetry to police; opens live audio bridge.

5. Scalability, High Availability (HA) & Security Specifications

A. Concurrency Control & High Availability

• **Redis Redlock (Distributed Locking):** Prevents race conditions where network latency causes two proximate workers to accept the exact same broadcast booking.
• **PostgreSQL Optimistic Concurrency Control (OCC):** Row versioning on the bookings table guarantees atomic state transitions, rolling back stale requests with a 409 Conflict.
• **High Availability Topology:** 1 Primary Write PostgreSQL node with 2 Read Replicas (supporting 15,000 req/sec) and automated multi-zone failover delivering 99.99% system uptime.

B. Zero-Knowledge Identity Protection & Regulatory Compliance

• **UIDAI Compliance:** Raw 12-digit Aadhaar numbers are never persisted. The platform generates an irreversible SHA-256(Aadhaar + Salt) token verified via Digilocker OAuth2.
• **EXIF Geofencing & Photo Proof:** Post-repair photos are inspected for live device GPS metadata. Uploads taken outside a 250m geofence of the booking location are rejected.
• **Code on Social Security (2020) Alignment:** Automatically routes the 1.5% cooperative cess into PMSBY accidental insurance and PACS welfare funds, fulfilling statutory compliance.

6. Nationwide Scaling Framework (4-Phase DPI Roadmap)

Phase & Timeline	Geographic Target	Workforce Target	Institutional Integration Milestones
Phase 1: Pilot Cluster (Months 1 – 6)	Delhi NCR / Capital Region	5,000 Technicians	Deploy in 10 urban labour federations. Validate fallback state machine, dual OTP, and SOS police link.
Phase 2: Tier-2/3 Expansion (Months 7 – 18)	25 Cities across 5 States	50,000 Technicians	Integrate with State Cooperative Boards. Launch cooperative hardware B2B network. Onboard 1,000+ NGOs.
Phase 3: Pan-India Integration (Months 19 – 36)	Nationwide (All States/UTs)	500,000+ Technicians	Full integration with Ministry's PACS Computerization Cloud & NCDC grants. Launch PSU CSR voucher portal.
Phase 4: National DPI (Months 37 – 60)	District & Block Clusters	2,000,000+ Guild Members	Capture institutional government maintenance contracts (Panchayats, Railways). Fulfill 'Sahakar Se Samriddhi'.

High-Level Design Certification: The Seva Setu enterprise architecture satisfies all technical, operational, and governance requirements for a sovereign Digital Public Infrastructure (DPI). It decouples worker earnings from corporate extraction, provides zero-downtime service fallback for citizens, and equips the Ministry of Cooperation with macro-level workforce planning intelligence.